



Diploma Thesis

Helikon

Literaturrecherche

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Statutory declaration

I declare that I have authored this thesis independently, that I have not used other than the declared sources and that I have explicitly marked all material which has been quoted either literally or by content from the used sources. I also used the following generative AI tools [e.g. ChatGPT, Grammarly Go, Midjourney] for the following purpose:

- ChatGPT: Research of scientific papers, sentence formulation, synonyms, brainstorming, diagram generation

Vienna, 24.10.2025

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Chapter 1

Literature Research

Artificial intelligence (AI) represents a key technological paradigm shift of the 21st century, enabling machines to perform tasks that previously required human cognition. Its development has led to significant progress in natural language processing, data analysis, and decision-making systems. In the field of education, AI supports teachers and learners through adaptive systems, automated feedback, and the intelligent organization of digital learning materials. By transforming how knowledge is processed and personalized, AI has become a fundamental driver for innovation in modern educational environments [1].

In spite of the current speed at which AI is being integrated into everyday life, its use in classical education systems remains limited. The lack of secure, domain-specific tools that assure full control over copyright and data privacy blocks teachers from effectively using such technologies in a classroom setting. The goal of Helikon is to address this issue by combining retrieval-augmented generation, vector databases and user-rights management to provide a reliable, adaptable and teacher-centered AI platform. This leads to the central question: Which vectorization and retrieval architectures provide an effective and secure foundation for integrating teacher-specific materials into large language model-based educational platforms?

The following sections cover general theoretical foundations and are followed by specific implementation concepts. The first section boils down the pedagogical and technological background of artificial intelligence in education. Next, the theoretical basis for knowledge representation and vectorization is discussed, followed by a summary of retrieval-augmented generation. Furthermore, the essential theory behind vector databases and information retrieval is presented, followed by an exploration of data-security and access control in educational AI systems. Finally, a synthesis and analysis of research gaps is provided.

1.1 Foundations of Artificial Intelligence in Education

Research in artificial intelligence applied to education (AIED) began in the 1970s with the development of rule-based intelligent tutoring systems (ITS), which attempted to model student knowledge and adapt instruction accordingly [2]. Since then, research has crossed several technological thresholds:

1. **Expert and rule-based systems:** predefined logic trees catered for specific subjects that aid in solving problems [3].
2. **Machine-learning approaches:** personalized exercises and feedback using data [2].
3. **Large language models (LLMs)** transformer-based architecture enabling dialogue, text generation and contextual reasoning [4]

From a pedagogical point of view, AIED benefits both the student and the teacher through adaptive learning, where it is possible to dynamically adjust difficulty levels based on performance [2]. Furthermore, automated assessment and overall teacher assistance like summarization and examples generation are additional reasons why AIED should be reinforced. However, several ethical and practical challenges arise:

- **Data privacy and copyright:** The teacher must remain in control over how their data are stored and processed.
- **Transparency and bias:** The behavior of models should be interpretable and not lean on certain biases that may be created by training data.
- **Pedagogical alignment:** The model should support rather than replace human instruction

To ensure proper functionality, AI-based learning systems rely on computation forms of knowledge. Text and educational content in general needs to be transformed into *embeddings*, allowing for contextual information retrieval. [5]

1.2 Knowledge Representation and Vectorization

Data quantity doesn't determine the effectiveness of AI models. Instead, it's how knowledge itself is represented and processed by the model [5]. This section introduces the concept of *knowledge representation* (KR), which details how information is structured for later reasoning.

1.2.1 Concept of Knowledge Representation

At its core, KR concerns itself with the formal structuring of information with the goal of creating optimized frameworks for AI's to reason, infer and learn from [2]. The information transformed ranges from abstract concepts to relationships and facts and the process itself serves as the foundation for all higher reasoning tasks, memory, decision-making, etc.

Two main paradigms of knowledge representation are currently relevant: symbolic and sub-symbolic [6]. The symbolic paradigm utilizes explicit and interpretable rules, logic statements and relationships. For example, knowledge graphs. An eminent drawback of such approach is the systems rigidity and inability to capture the context-rich nature of human language.

In contrast, the sub-symbolic paradigm represents knowledge through distributed numerical encoding. Information is encoded in continuous vector spaces that capture the essence of human language. This approach provides the basis for the development of embedding-based language models, which map linguistic or conceptual entities into dense numerical representations [5]. The paradigm has been further enhanced by transformer-based architectures such as **BERT** which introduced context-depen-

dent representations, allowing words to take on different meanings depending on its surroundings [5]. **Sentence-BERT** expanded this principle to entire sentences and documents [7],

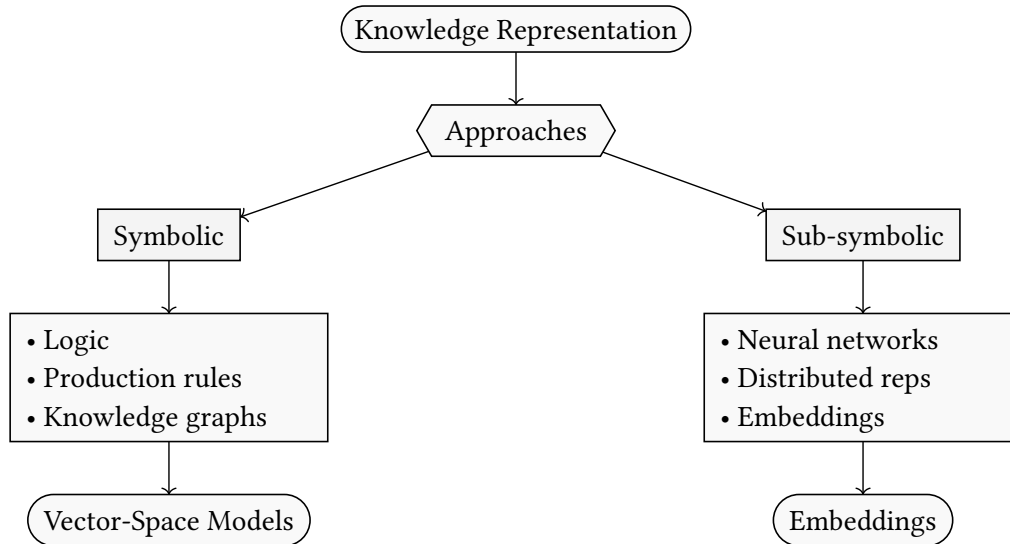


Figure 1.1: Knowledge Representation – Symbolic vs. Sub-symbolic [8]

The relationship between symbolic and sub-symbolic paradigms is illustrated in Figure 1.1, which underlines the fact that both can be seen as complimentary. While symbolic models offer interpretability and structure, sub-symbolic models focus on flexibility and contextual understanding [6]. As a side note, vector-space models (VSM) found in the symbolic branch are not to be confused with vector-spaces described in the sub-symbolic section. They are independent concepts, VSMs are vectors derived from explicit, count-based statistics and vector spaces are learned, dense and contextualized vector encodings [7].

1.2.2 Word and Sentence Embeddings

Using the sub-symbolic paradigm as a basis, **embeddings** represent an essential mechanism used by AI systems to encode meaning. Using this approach, words, sentences and documents are transformed into a continuous vector representation that encodes the context-rich nature of human language [7]. As a consequence of transforming language into vectors, the *vectorization* process allows for numerical measurement of concept similarity.

The similarity between two embeddings, for example between a query q and a document d_i is commonly computed through **cosine similarity**, which measures the angle between two vectors [9]:

$$\text{sim}(q, d_i) = \cos(q, d_i) = \frac{q \cdot d_i}{\|q\| \|d_i\|} \quad (1.1)$$

A higher value indicates stronger alignment between the two representations. To transform similarity values into normalized relevance probabilities, the **softmax function** is frequently used [10]. Using a temperature parameter τ :

$$P(d_i | q) = \frac{e^{\tau \cos(q, d_i)}}{\sum_j e^{\tau \cos(q, d_j)}} \quad (1.2)$$

The mentioned mathematical principles allow embeddings to be used for retrieval, clustering and semantic search. They allow AI systems to generalize and operate on conceptual relationships, which is the basis for retrieval-augmented generation (RAG) frameworks, discussed in the following sections [4]. In RAG, embeddings are fetched and integrated dynamically into LLMs.

1.2.3 Evaluation Criteria for Embeddings

The quality and value of an embedding model can be assessed through several important criteria that reflect the model's ability to capture and retrieve information effectively. The general goal is balancing the right trade-offs between accuracy, computational cost and domain applicability. The following aspects are most relevant in the context of educational AI systems.

- **Semantic Accuracy:** A measure of how well embeddings preserve meaning and context after the transformation. High semantic accuracy means that semantically similar concepts, like *question* and *inquiry*, are represented by neighboring vectors in the embedding space [7].
- **Dimensionality:** Refers to the amount of dimensions in the vector representation. Higher dimensionality can describe data better and capture more subtle differences between concepts but also drastically increases memory and time complexity.
- **Domain Specificity:** General-purpose embeddings trained using broadly sourced text data may underperform when presented with specialized domains such as mathematics. It is therefore essential to keep in mind that fine-tuning on teacher-selected materials can improve contextual specificity [10].
- **Computational Efficiency:** Measures the overall performance of embedding-based systems, including speed, scalability and memory consumption. While the previously discussed lower-dimensional embeddings can improve efficiency, implementations such as vector indexing, batching and hardware acceleration also play a significant role in performance [11].
- **Robustness and Transferability:** Embeddings should be able to generalize across concepts and remain functional when presented with noisy input, ensuring consistent performance in the real world.

Together, these aspects provide solid criteria for evaluating embedding models both in terms of their numerical performance and their pedagogical applicability in real-world scenarios.

1.3 Retrieval-Augmented Generation (RAG)

While large language models (LLMs) have access to extensive general knowledge, they are limited by static data and potential inaccuracies in them [12]. To solve this eminent constraint, retrieval-augmented generation (RAG) is used to provide a system for retrieving up-to-date and context-relevant data during the generation process. This section focuses on the theoretical foundation, system architecture and practical challenges of RAG in educational AI applications.

1.3.1 Theoretical Framework

Retrieval-augmented generation combines information retrieval and generative language modeling to form a unified framework that improves interpretability. Rather than forming a sole dependence on the internal model's parameters, RAG has access to an external storage, often a **vector database** (discussed in Section 1.4), to retrieve relevant contextual information used to improve response quality.

Conceptually, RAG consists of two modules:

- **Retriever:** Transforms a user query into an embedding vector and performs a semantic search for similar database documents.
- **Generator:** Produces a response based both on the query and the retrieved contextual information from the retriever. [12]

Using Equation (1.2), a retriever can compute the probabilistic relevance between a user query and documents stored in the database. The result is then passed down to the generator, which produces a response y based on both the input x and the documents D

$$P(y | x) = \sum_{d \in D} P(y | x, d)P(d | x) \quad (1.3)$$

This equation illustrates how the retrieved information d influences the process of generating the response by weighing each document's contribution to the final output. [4]

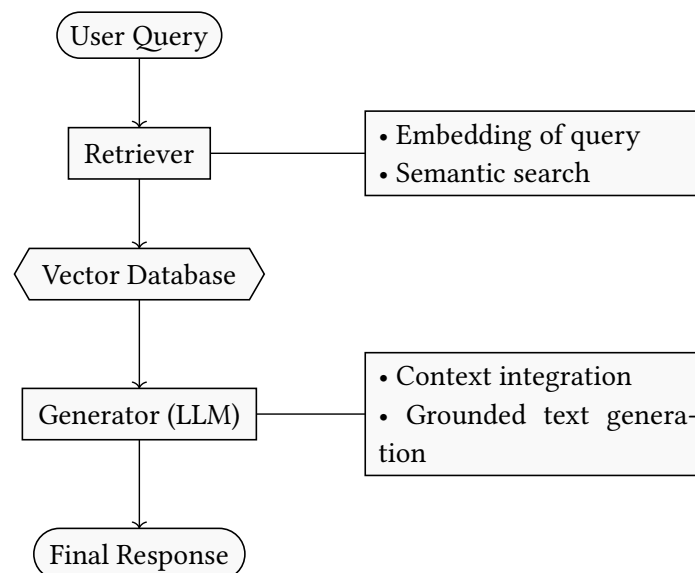


Figure 1.2: Retrieval-Augmented Generation (RAG) Architecture [13]

As shown in figure Figure 1.2, the retriever transforms the user query into an embedding and performs a semantic search identifying similar documents within the vector database. The information is then passed down to the generator, which turns it into a proper response.

1.3.2 Research Findings on RAG Systems & Open Research Challenges

Research indicates that retrieval-augmented generation helps with factual grounding on knowledge-dependent tasks. Key findings relevant to AI in the educational field include:

- **Factuality & Evidence use:** Having a basis in form of a vector database for contextual knowledge reduces hallucinations and promotes source-based answers [4], [12].
- **Domain adaptation:** Teacher are now able to personalize the domain specificity of a model by injecting curated materials without the need to fully retrain the base model [4], [10].
- **The importance of retrieval:** Overall output quality depends mostly on retriever recall and index design, not solely on the LLM [10], [11].

- **Scalability of similarity search:** Billion-scale retrieval with GPU acceleration is possible and would enable access to large datasets with a low latency. [9]
- **Safety hooks:** It is feasible to integrate policy filters into the retrieval stage to ensure response alignment with organizational rules [14].

In spite of a number of benefits, several issues remain for classroom-tuned RAG's:

- **Context windows limits:** Retrieval output needs to be properly chunked to fit the model's input budget. This topic remains an active area of research [12].
- **Retrieval quality:** Keeping a high recall and omitting off-topic contextual drifts are areas that still need refinement [12], [10].
- **Indexing trade-offs:** ANN choices balance latency, recall and cost. Choosing specific trade-offs in the educational field is dataset-specific [11], [9].
- **Attribution:** Consistent citation across both retrieval and generation stages is required to ensure that every claim bases itself on verifiable source [12].
- **Security, privacy & access control:** It is important to prevent the leakage of confidential materials and enforce permission calls for teachers that result in policy checks at retrieval time (e.g. RBAC). Ideally, such policy checks should be directly integrated into the RAG loop [14], [15].

These findings and problems motivate a shift towards vector databases and retrieval theory, the focus of the next section.

1.4 Vector Databases and Information Retrieval Theory

Retrieval systems are responsible for determining how efficiently information originating from external sources is captured and are the core of RAG's. The effectiveness of an LLM depends on the quality and performance of its retrieval process, which vector databases try to address. They store semantic embeddings and enable efficient similarity searches across large datasets. The following section explains the theoretical foundations of retrieval, the structure of vector databases and performance considerations.

1.4.1 Theoretical Background of Information Retrieval

The main focus of information retrieval (IR) lies in identifying useful information from a large document database based on a query. Classical IR relies on *lexical matching* techniques, which used frequency and occurrence probability to estimate the importance of documents. One of the most influential models was **BM25**, which predict document d 's importance to a query q using term frequencies and normalization parameters [9]:

$$\text{score}(q, d) = \sum_{t \in q} \text{IDF}(t) \cdot \frac{f(t, d) \cdot (k + 1)}{f(t, d) + k \cdot \left(1 - b + b \cdot \frac{|d|}{\text{avgdl}}\right)} \quad (1.4)$$

Where:

- $f(t, d)$ - frequency of term t in doc d
- $|d|$ - doc length
- avgdl - average doc length
- $\text{IDF}(t)$ - inverse doc frequency of term t
- k and b - scaling constants that control the importance of the frequency and doc length terms.

While BM25 are effective at capturing word importance, their limitation lies in the surface-level similarity, that relies on literal keyword overlaps between the query and the document. The consequence of such limitation is the failure to identify relationships based on meaning.

Modern IR systems use embedding-based retrieval, where both documents and queries are used as vectors in a multidimensional space. This results in a geometric computation, removing the need for word overlap and enabling the retrieval of similar materials that don't have identical wording [10], [11].

This expansion of retrieval theory provided a basis for **vector databases**. They allow embeddings to be stored, indexed and compared while staying efficient and fast.

1.4.2 Conceptual Structure of Vector Databases

Vector Databases (VDBs) are a foundation for embedding-based retrieval systems. Their primary task is the organization of high-dimensional vector representations of data. In contrast to relational databases that use key matching, vector databases use the *Approximate Nearest Neighbor* (ANN) search to identify similarities based on geometry [11].

A vector database consists of three layers [11]:

- **Storage layer:** Manages and persists embedding vectors and their metadata
- **Indexing layer:** Organization of vectors into optimized structures for efficient searching, such as clustering or graph-based networks.
- **Retrieval layer:** Ranking, querying and filtering results based on distance metrics.

There are several indexing strategies for achieving scalable performance [9]:

- **Flat index:** An exhaustive search involving all vectors in a database. This approach provides maximum accuracy at the cost of a high computational load.
- **Inverted file index (IVF):** Uses k-means clustering to partition the vector space, allowing for efficient searching by only focusing on relevant clusters
- **Hierarchical Navigable Small World (HNSW):** Utilizes graph-based hierarchy that uses edges to connect neighboring vectors. This approach ensures efficient traversal for larger sets of data.

Apart from indexing, **contextual isolation** is a key requirement for scalable architectures in educational retrieval systems. Since there exist separations on grade levels, subject levels, etc., it is insufficient to use a single shared index. Rather, metadata tagging is used to partition vectors within the same database instance. Each embedding should be associated with categorical tags, such as “teacher”, “ID”, “grade” and “module name”. During similarity search, they act as filters to ensure unrelated materials that are semantically similar aren't being used. Such tag-based systems support multi-tenant architecture, meaning each teacher manages several vector spaces at once [10], [11].

1.4.3 Real-Time Data Flow in Retrieval Systems

While Section 1.3 introduced retrieval-augmented generation from a functional standpoint, the underlying retrieval introduces specific considerations regarding *latency*, *scalability* and *efficiency*. An important detail to consider is that in real-time environments, the perceived intelligence of an AI system, relies on the responsiveness of the retrieval pipeline.

During a typical query cycle, three stages can be identified:

1. **Query embedding:** The text input is transformed into an embedding. It is then further pre-processed using, for instance, normalization and batching to allow parallel handling of multiple queries. Batching additionally reduces overhead and improves hardware utilization [10].
2. **Vector search & Filtering:** The embedding enters the index structure, where methods discussed in Section 1.4.2 are applied, including ANN techniques such as HNSW and IVF which directly determine the speed and precision of retrieval [9]. Additional steps, such as the previously discussed *tag-based isolation*, restrict the search to relevant clusters, ensuring contextual separation.
3. **Ranking:** Similarity scores and contextual metadata are used to rank the retrieved candidates. Ranking computations combine both geometric distance and statistical relevance, often including factors like recency or pedagogical reference. The final list is then typically passed down to the RAG generator.

From a theoretical perspective, system performance is dependent on:

- **Computational complexity:** ANN traversal cost is increased sub-linearly with index size when using clustering or hierarchical methods. Nonetheless, dimensionality and precision thresholds still pose the practical limits of real-time operations [9].
- **Latency propagation:** Each retrieval stage adds to the total response time. Even when index structures are optimized, network communication, data transfer and vector serialization factors may become primary sources of delay [11].

To achieve noteworthy scalability in educational contexts, a balance between precision and throughput must be maintained. When multiple teachers try to access separate tagged datasets concurrently, the retrieval operations can't exceed latency thresholds. Vector database frameworks have been proposed, where computational resources are dynamically allocated according to active tags and working implementations of such multi-tenant systems already exist [10].

1.5 Data Security and Access Control in Educational AI Systems

Data security forms the foundation of responsible AI systems in education. Since educational fields require systems to process personal and pedagogical data, maintaining integrity, confidentiality and accessibility control is essential. The role of access control models is to shape how users interact with system resources and how data boundaries are enforced [15], [14]. The following section discusses theoretical access control approaches alongside ethical and legal principles that are the base of secure AI systems, especially in the educational field.

1.5.1 Theoretical Models of Access Control

Access control models are responsible for laying the foundation of the structure in which users interact with system resources and determine permission boundaries. Two primary paradigms emerge in literature: **Role-Based Access Control** (RBAC) and **Attribute-Based Access Control** (ABAC).

RBAC organizes permissions based on roles rather than individual users. A role defines a set of permissions that allow actions on certain resources. Users are assigned roles, and roles link to permissions, creating a clear and maintainable structure [15]. This model is well suited for educational institutions, where there are hierarchical roles with clear access rights, such as *student*, *teacher* or *administrator*. Advantages of RBAC include simplified administration and policy enforcement clarity, though it can become too rigid in situations where context is important for access.

ABAC extends RBAC by basing its decisions on certain attributes assigned to users, rather than fixed roles. These attributes may include user identity, type of resource or time of access. As a result, ABAC provides a more dynamic approach that is becoming increasingly important with the rise of AI-driven systems where contextual factors, like grade level or learning module, are relevant factors in granting retrieval permissions.

As a result, both RBAC and ABAC are relevant for constructing AI-based multi-user educational systems. However, RBAC favors administrative simplicity, while ABAC emphasizes adaptability. Hybrid implementations often combine the RBAC and ABAC paradigms to achieve an optimal balance between context flexibility and management efficiency. They currently represent the most suitable solution for access control in the educational AI field. For example, a teacher may be given a general role that grants access to a specific subject area, but specific retrievals could still depend on attribute checks such as the grade level or module tag.

1.5.2 Ethical and Legal Foundations

The integration of AI into educational fields requires not only technical analysis but also compliance with ethical and legal standards. European data protection regulations, such as the **General Data Protection Regulation** (GDPR), provide a legal framework for processing personal data within an educational institution [16]. GDPR underlines lawfulness, purpose limitation and data minimization, requiring data collection on a minimal scale.

In AI systems, these principles translate to transparency of how data are processed and accountability for automated decision-making. Access control mechanisms, discussed in Section 1.5.1 contribute to compliance by allowing only authorized users to retrieve or modify protected resources [14].

Beyond legal compliance, trust and explainability are ethical foundations of responsible AI-systems. Trust is established when teachers know that the system they're using is secure and reliable, while explainability is achieved by ensuring that access logic of the model remains understandable.

Consequently, the principles of data ethics and legal compliance converge with access control models to design secure and responsible frameworks for AI in education. A secure educational system should therefore enforce legal, ethical and technical rules to enable a sustainable AI adoption in schools and universities.

1.6 Synthesis and Research Gaps

The literature reviewed throughout this chapter establishes a coherent theoretical foundation for a secure, responsible and effective way of AI integration into educational systems. The examined sources highlight several themes: how important structured knowledge is, the role of vector databases in semantic retrieval, the reasoning enhancement of using retrieval-augmented generation and the need for secure access control protecting user data.

Despite a substantial amount of progress, several major gaps in research remain:

- **Domain specificity:** Current embedding and retrieval based models are primarily trained on open-domain data. Their transferability to educational contexts remains largely unknown [7], [10].
- **Policy-aware retrieval:** Existing theoretical models rarely tackle the challenge of ensuring fine-grained, role-based information isolation within multi-tenant educational environments.

- **Pedagogical transparency:** Ethical discussions often underline the importance of explainability and trust, yet few works provide concrete foundations for evaluating interpretability within educational AI systems.
- **Scalability under legal constraints:** Although vector databases offer efficient large-scale retrieval discussed in the sections before [9], [11], their performance and compliance under regulations such as GDPR [16] remain unexplored.

The Helikon project aims to address these limitations by uniting retrieval-augmented generation, vector-based data storage and dynamic access control into a cohesive architecture. This structure will allow teachers to integrate their own materials while preserving both privacy and semantic precision. By also including modular retrieval, tag-based isolation and transparent policy enforcement, Helikon turns the theoretical principles discussed into a practical, teacher-centered system design.

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List of Prompts

- [1] OpenAI, “PROMPT, ChatGPT GPT-5. Write a factual and neutral introductory paragraph for a diploma thesis in computer science about the general significance and role of artificial intelligence in education. The paragraph should be formal, academic in tone, and suitable as the opening of a literature review chapter.” Oct. 05, 2025.
- [8] OpenAI, “PROMPT, ChatGPT GPT-5. Erstelle mit Fletcher ein Diagramm, das den Unterschied zwischen symbolischer und sub-symbolischer Wissensrepräsentation zeigt. Ausgangspunkt ist Knowledge Representation, das zu Approaches führt und sich in zwei Zweige teilt: Symbolic mit Logic, Production Rules und Knowledge Graphs sowie Sub-symbolic mit Neural Networks, Distributed Representations und Embeddings. Von dort gehen Pfeile weiter zu Vector-Space Models beziehungsweise Embeddings. Verwende schlichte Formen wie pill, hexagon und rect, dezente Grautöne (luma 96–98 Prozent), klare Abstände und setze die Beschriftung als Caption Knowledge Representation — Symbolic vs. Sub-symbolic. ” Oct. 24, 2025.
- [13] OpenAI, “PROMPT, ChatGPT GPT-5. Erstelle mit Fletcher ein Diagramm, das die Architektur von Retrieval-Augmented Generation darstellt. Beginn mit dem Knoten User Query, der über Pfeile zu Retriever, Vector Database, Generator (LLM) und schließlich Final Response führt. Neben Retriever soll ein Knoten mit den Punkten Embedding of query und Semantic search stehen, neben dem Generator ein weiterer mit Context integration und Grounded text generation. Verwende einfache Formen wie pill, rect und hexagon, dezente Grautöne (luma 96–98 Prozent), gleichmäßige Abstände und setze die Beschriftung als Caption Retrieval-Augmented Generation (RAG) Architecture. ” Oct. 24, 2025.

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